

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks available |     |     |       |       |      |
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| 5        |  |  |  | <p>Indicative content</p> <p><b>Conditions required for germination</b></p> <ul style="list-style-type: none"><li>• Water + oxygen + suitable temperature</li></ul> <p><i>Water</i></p> <ul style="list-style-type: none"><li>• Cotyledons swell/ testa softens</li><li>• Transport – dissolve substances/fluid medium for enzymes</li></ul> <p><i>Oxygen</i></p> <ul style="list-style-type: none"><li>• (Aerobic) respiration –</li><li>• energy/ATP for metabolism</li></ul> <p><i>Suitable temperature</i></p> <ul style="list-style-type: none"><li>• Speeds up rate of diffusion</li><li>• Increases enzyme activity</li></ul> <p><b>Germination of Peanut</b></p> <ul style="list-style-type: none"><li>• Non endospermic/endosperm absorbed(into cotyledons)/no endosperm</li><li>• Amylase digest starch in cotyledons to maltose</li><li>• {Proteins/fats} broken down into {amino acids/fatty acids and glycerol}</li><li>• Move to {plumule/radicle/meristem/sink}<br/>(Can apply to peanut or barley)</li><li>• For {mitosis/growth/cell division}<br/>(Can apply to peanut or barley)</li></ul> <p><b>Germination of barley</b></p> <ul style="list-style-type: none"><li>• Endospermic/endosperm present</li><li>• Starch/ proteins/fats in endosperm</li><li>• Embryo produces {gibberellic acid/gibberellin}</li><li>• Gibberellic Acid {moves into/stimulates} aleurone layer</li><li>• Gibberellic Acid causes enzymes to break down protein into amino acids</li><li>• Amino acids used to synthesise enzymes such as amylase</li><li>• Amylase breaks down starch in the endosperm into maltose</li></ul> | 3               | 6   |     | 9     |       |      |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |     |     |       |       |      |
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|          |  |  |  | <p><b>7-9 marks</b><br/>Indicative content<br/>All three parts covered in details<br/><i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i></p> <p><b>4-6 marks</b><br/>Indicative content<br/>Two parts covered in detail<br/><i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i></p> <p><b>1-3 marks</b><br/>Indicative content<br/>Only one part of the question is addressed.<br/><i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i></p> <p><b>0 marks</b><br/><i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i></p> |                 |     |     |       |       |      |
|          |  |  |  | Question 5 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3               | 6   | 0   | 9     | 0     | 0    |